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Polarization-sensitive laser feedback interferometry for specular reflection removal

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Specular reflection from the surface of targets or prepared specimens represents a significant problem in optical microscopy and related optical imaging techniques as usually the surface reflection does not contribute to the desired signal. Solutions exist for many of these imaging techniques; however, remedial techniques for imaging based on laser feedback interferometry (LFI) are absent. We propose a reflection cancellation technique based on crossed-polarization filtering that is tailored for a typical LFI configuration. The technique is validated with three experimental designs, and a significant improvement of about 40 dB in the ratio of the diffuse and specular LFI signal is observed. Applications of this principle extend from specular reflection removal to characterization of target materials in industrial to biomedical domains.

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1. INTRODUCTION

Surface reflection from samples or transparent mounts where, for example, air-glass interfaces exist is usually a source of contaminating signal prominent in reflective illumination microscopy, when one is interested in the signal from within the samples. Techniques to reduce this specular reflection include coatings to optical elements and liquid immersion techniques. These approaches, used individually or in combination, can successfully address this problem in conventional optical microscopy. However, there are numerous applications where anti-reflection coatings are not a practical solution. For example, surface reflection is a major issue in *in vivo* skin imaging, and clearly an alternative solution is required. In this paper, we use cross-polarization filtering with laser feedback interferometry (LFI) imaging to effectively suppress the specular reflection in a contact-free way, obviating the need for anti-reflection coatings. LFI is an interferometric laser technique based on the self-mixing effect [1–3], particularly well-suited to reflection-mode sensing. With the growing uptake of LFI in industrial and biomedical imaging [4–7], a solution to the target surface reflection contamination of signal is desirable.

Polarization-sensitive sensing and imaging have been suggested in conjunction with other optical techniques such as parallel and perpendicular polarization photography [8], polarized light camera imaging [9], crossed-polarization filtering [10], and polarization-sensitive optical coherence tomography [11]. In the context of LFI, signals from birefringent orthogonal-polarization dual-frequency lasers [12,13] and bistable polarization vertical-cavity surface-emitting lasers (VCSELs) [14] have been investigated, and polarization active components have been employed to stop parasitic reflection from the interfaces to enter the laser cavity and contribute to the signal [15]. However, such polarization-sensitive configurations have not been explored in LFI imaging applications for the purpose of surface specular reflection suppression. Using crossed-polarization filtering, we distinguish between specular and diffuse components of the backscattered beam and effectively suppress the specular part. This technique is able to improve the versatility of the existing LFI systems in numerous ways. It enhances the LFI signal to background ratio when signal source is situated within a sample or when sharp specular reflection on the surface drowns out the signal. Polarization-sensitive LFI can provide a simple platform for an affordable and compact tool for biomedical imaging applications [7], to obtain the signal from deep within tissue and suppress the front-surface reflection. The advantage over similar modalities is a relaxed depth of field constraint as the front surface signal can be incorporated in the reflected signal if it is subsequently nulled and offers the ability to use lower numerical aperture beams with longer depths of field. Such a system can avoid immersion lenses or index matching gels for specularity removal. Additionally, the instrumentation required for LFI flow Doppler studies in microfluidic channels [16] can be minimized if the need for index matching liquids or high numerical aperture beams is removed. A series of three experiments will be presented to validate the approach. The first two will quantify the benefit of reducing specular reflection in the context of surface imaging, while the third will demonstrate the advantage if this approach is used for in-depth interrogation of samples in microfluidic channels.

After presenting a theoretical framework for the technique in Section 2, the idea is examined experimentally through different experiments in Section 3. Section 4 discusses the results, and Section 5 provides a conclusion for the paper.

2. LASER FEEDBACK INTERFEROMETRY AND POLARIZATION CONTROL

LFI is based on the self-mixing effect: the emitted light from a laser is partially reflected from an external target and re-enters the laser cavity where it interferes with the intracavity fields and creates an interferometric signal [3]. It is a compact technique in which the laser acts as both transmitter and receiver of the beam, used in a wide range of applications [17–21]. To make it polarization-sensitive, we added a polarizer and a quarter-wave plate (QWP) to a confocal LFI setup. Figure 1 illustrates the configuration for the proposed polarization-sensitive LFI system. In Fig. 1, the laser beam from a VCSEL passes through a polarizer oriented parallel to the plane of the illuminating beam and a QWP before being incident upon a sample. The VCSEL, which was used in the experiments, had a linearly polarized beam; however the technique works for nonpolarized light sources as well. The input polarizer, which has a vertical orientation, performs two tasks: passing the vertical component of the input beam to the target and suppressing the cross-polarized component of the reflected beam on its way back to the laser.

Figure 2 shows a detailed graphical view of the changes in the polarization state of a beam in a polarization-sensitive LFI

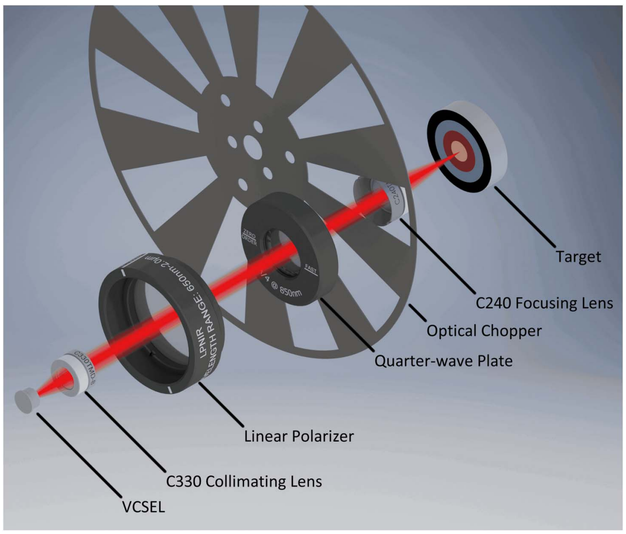

Fig. 1. Configuration of a polarization-sensitive LFI system. Lenses are used to collimate and focus the beam, and an optical chopper modulates the beam. A polarizer and quarter-wave plate (QWP) control the polarization of the incident beam and filter out the reflected beam.

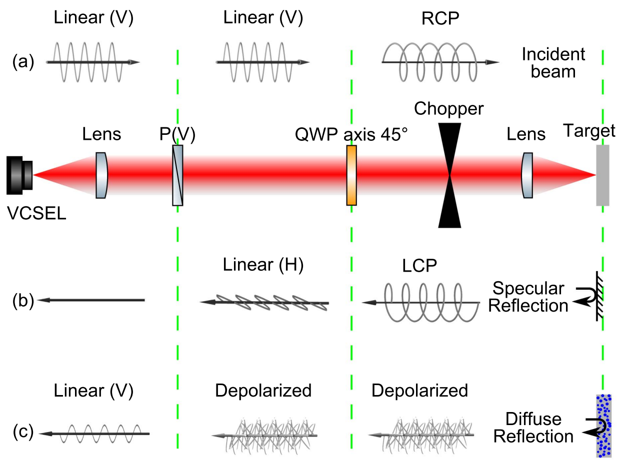

Fig. 2. Schematic diagram of a polarization-sensitive LFI system and changes to the polarization state of the beam passing through the components. Three green broken lines are the locations where the beam's polarization state may change. Wave diagrams show the polarization conversion of the (a) incident, (b) specularly reflected, and (c) diffusely reflected beams. Optic axis of the QWP is at 45° to the linearly polarized beam. Vertical (V), horizontal (H), polarizer (P), right-hand circularly polarized (RCP), left-hand circularly polarized (LCP).

for a specific state when the QWP optic axis is at 45° to the incident linearly polarized beam. Three green broken lines in this figure show the locations where the polarization state of the beam may change. Depending on the orientation of the QWP, the polarization state of the beam may change to right-hand circularly polarized (RCP) or left-hand circularly polarized (LCP), or remain linear. Linear polarization of the beam is preserved when fast or slow axes of the QWP are oriented parallel to the direction of the incident linearly polarized beam, whereas the polarization state changes from linear to circular when the QWP optic axis is at 45° to the orientation of the incident linearly polarized beam [as illustrated in Fig. 2(a)]. Placing the QWP's slow axis at 45° to the linearly polarized beam results in an RCP beam [Fig. 2(a)], whereas setting the QWP's fast axis at 45° to the linearly polarized beam results in an LCP beam. The change in the polarization state from linear to circular is due to a 90° phase shift between the two perpendicular electric field components of the incident beam.

Other components in the setup such as lenses and an optical chopper do not change the polarization state of the beam. The only other phenomenon that can change the polarization state of the beam in this setup is the reflection from the target. Upon reflection, a polarized beam may remain polarized or become partially or fully depolarized. In the case of the beam that remains polarized, polarization may be altered from one form to another. Any change in the polarization state of the reflected beam depends on the nature of the reflecting target. In this context, one can categorize different types of reflecting targets into two main groups: specular targets and diffuse targets. A specular target generates a specular reflection [Fig. 2(b)] associated with a glossy appearance, whereas a diffuse target generates a diffuse reflection [Fig. 2(c)] associated with a matte appearance. A coherent process in the reflection retains the polarization as in the case of a specular reflection [Fig. 2(b)], while an incoherent process scrambles the polarization as in the case of a diffuse reflection [Fig. 2(c)]. Diffuse reflection usually occurs when the beam penetrates the surface and undergoes multiple scattering within the sample [Fig. 2(c)], while specular reflection occurs on the surface due to refractive index mismatch with equal incidence and reflection angles [Fig. 2(b)]. A linearly polarized beam retains the same polarization state upon normal specular reflection, while the handedness of a circularly polarized beam becomes reversed upon such a reflection [RCP in Fig. 2(a) changes to LCP in Fig. 2(b)]. This reversal in handedness of a circularly polarized beam on reflection is the key to crossed-polarization filtering as the reversed circularly polarized beam changes into an orthogonal linearly polarized beam after passing through the QWP on its way back which will then be blocked by the vertically oriented polarizer [Fig. 2(b)].

As explained briefly before and shown in Fig. 2, depending on the QWP angle, two operational modes can be considered for a specular reflection. In the first mode, when the QWP optic axis is at 45°, a circularly polarized beam reflects back from a specular target with a reversed handedness. As it passes through the QWP on its way back, the reversed circular beam converts into a horizontal linearly polarized beam with respect to the original beam orientation. Horizontal polarized beams will be filtered out by the vertical polarizer meaning that in this mode, specular reflection does not contribute to the signal. The second mode occurs when the QWP angle is at 0°. In this mode polarization remains linear upon specular reflection and contributes to the signal after passing through the other components. In the case of diffuse reflection, regardless of the incident polarization, the reflected beam is depolarized and passes through the QWP with no change in its polarization state. However, the horizontal component of the depolarized beam will be blocked by the vertical polarizer, and only half of the beam with parallel polarization passes through and contributes to the signal. Therefore when the QWP optic axis is at 45°, this method is capable of separating specular and diffuse parts of reflected beams and filtering out of the specular part. We also present a matrix model to investigate the deformation of the polarization state throughout the system based on Stokes parameters and Mueller matrix formulation (see Appendix A). Specular reflection is undesirable in many applications as it originates from the surface and buries the information from within the samples. Such a technique can be of great importance in noninvasive biomedical imaging systems when the signal from within the tissue is the main source of information.

3. EXPERIMENTS AND RESULTS

We examined the idea experimentally in three sets of experiments. First, we measured the LFI signal from standard specular and diffuse targets along one and two dimensions while sweeping the QWP angle. This experiment was conducted as a proof of concept and investigation of the effect of the QWP angle (system's degree of freedom) on the LFI signal. Then, we imaged a piece of specular plastic with engraved diffuse letters on it at a fixed QWP angle associated with the maximum filtration. In this experiment, further investigation of the level of filtration between specular and diffuse parts of the LFI signal was performed using a target with both types of surfaces at two QWP angles of 0° to 45°. Finally we scanned a microfluidic channel embedded in a silicone layer containing a fluid as a phantom for biological microcirculation. In this experiment, the main source of information (channel) is situated beneath a specular surface, and it is shown how this technique can be used to image biological tissue microcirculation without the need for index matching liquids, immersion lenses, or high numerical aperture beams.

Figure 1 illustrates the experimental setup. In this setup we used an 850 nm VCSEL (Litrax Technology Co., Ltd.) operating at 3.35 mA, just above the threshold current and temperature controlled at 35°C to obtain the best LFI signal [22]. The beam was collimated (using a C330TME-B, Thorlabs, Inc. lens) and focused (using a C240TME-B, Thorlabs, Inc. lens) on the target. The focusing lens had an 8 mm focal length and numerical aperture of 0.5. To control the incident polarization a polarizer (Melles-Griot Inc.) and a QWP (Melles-Griot Inc.) were placed in the path of the beam, with an extinction ratio of about 10^6. To perform imaging, targets were placed on a three-axis translation stage (Zaber Technologies Inc.) and raster scanned in front of the perpendicular laser beam when positioned at the focal plane of the beam. Signal sensing was performed point by point, and the image was constructed as a combination of the points. The LFI signal was acquired directly from the voltage of the VCSEL's terminals and fed into a computer using a data acquisition card. The changes in the voltage of the VCSEL's terminals are proportional to the changes in the VCSEL's optical power, caused by the self-mixing effect. To extract the voltage signal after an AC coupled amplification stage, the laser beam was modulated by an optical chopper at the frequency of 800 Hz, and the confocal reflectance signal was calculated as the difference between the two levels of signal at this frequency [23].

A. Quarter-Wave Plate Rotation

In the proposed polarization-sensitive LFI system, the angle of the QWP's optic axis with respect to the incident linearly polarized beam is an independent variable that can be controlled to separate the two components of the signal. To measure the suppression ratio or level of isolation between specular and diffuse reflections, we measured the LFI signal from specular and diffuse surfaces while rotating the QWP. Figure 3 shows

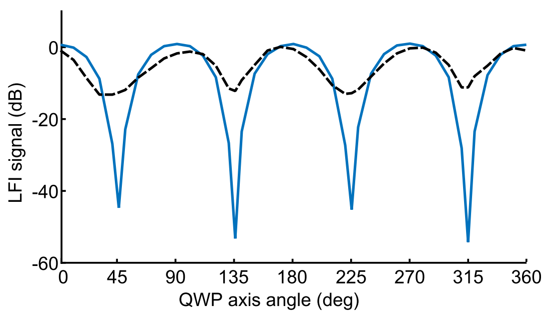

Fig. 3. LFI signal from mirror (blue line) and Spectralon (broken black line) plotted against QWP angle from 0° to 360°.Mirror is a specular, while Spectralon is a diffuse target.

the LFI signal in decibels from the surfaces of a mirror and Spectralon (Spectralon Diffuse Reflectance Standard, Labsphere Inc.) plotted against the angle of the QWP's optic axis to the orientation of the incident linearly polarized beam. The mirror has a specular surface, whereas Spectralon has a diffuse surface. Spectralon is a material that shows a good diffuse behavior and is used as a standard to create a Lambertian reflectance profile. Signals in Fig. 3 were normalized to their maximum values to have the same maximum level of signal (0 dB) for both surfaces. As explained in Section 2, only half of the electric field or a quarter of the optical power passes the system in the case of an ideal diffuse surface and there is no attenuation. That is a drop of at least 6 dB in the LFI signal from a diffuse surface expected when the QWP is at 45°. However, we observed drops of about 13 dB in the LFI signal from the surface of the Spectralon. Larger drops in the signal level could be caused by nonzero attenuation and imperfections in the alignment and the Lambertian profile of the reflection. We also used a focused beam setup, and there are difficulties in keeping a tiny focal point at the same position on the surface as the QWP rotates. Signal drops of about 45 dB (for QWP at 45° and 225°) and 55 dB (for QWP at 135° and 315°) can be observed in the case of the mirror. It means a suppression ratio of at least 30 dB (for QWP at 45° and 225°) and 40 dB (for QWP at 135° and 315°) can be observed in the level of the specular signal with respect to the diffuse signal. For instance, for the QWP angle of 45° to the linearly polarized beam, a separation larger than 30 dB can be achieved between the power of the two signal components. Therefore this scheme can provide a high level of specular filtration.

We also examined the suppression of the specular signal in an imaging application by imaging a piece of white paper next to a glass cover slip while sweeping the QWP angle. Reflection of a polarized beam from white paper is partially depolarized, whereas reflection from a glass cover slip retains polarization. Results are shown in Fig. 4 for the QWP angles of −33° to 137° at 10° pitch. This range covers the QWP angles of 45° and 135° where specular reflection is filtered out. The scan area is 2 mm × 2 mm and paper and glass are depicted at the top and bottom of the images in Fig. 4, respectively, as annotated in the first image. It is observed that the QWP angle does not greatly affect the image from the white paper, but the cover slip image is completely faded out when the QWP angle is close to 45° or 135°. Fringes on the cover slip images appeared because its surface is tilted with respect to the perpendicular laser beam. The changes in the orientation and density of the fringes can be caused by the slight misalignment in the optical system and strain on the cover slip.

B. Laser Engraved Letters

We used a piece of 6 mm thick acrylic plate with laser engraved letters on its surface as a sample consisting of both specular and diffuse surfaces. A laser cutter was used to engrave two letters (UQ) on the surface of the acrylic. The power of the laser was adjusted to a low level to create shallow letters on the surface with depths of about 50 μm. This way, in addition to the acrylic surface (top surface) that looks glossy, there are the engraved areas on the surface (bottom surface) that look matte. The laser evaporated the acrylic to create the bottom surface of the letters.

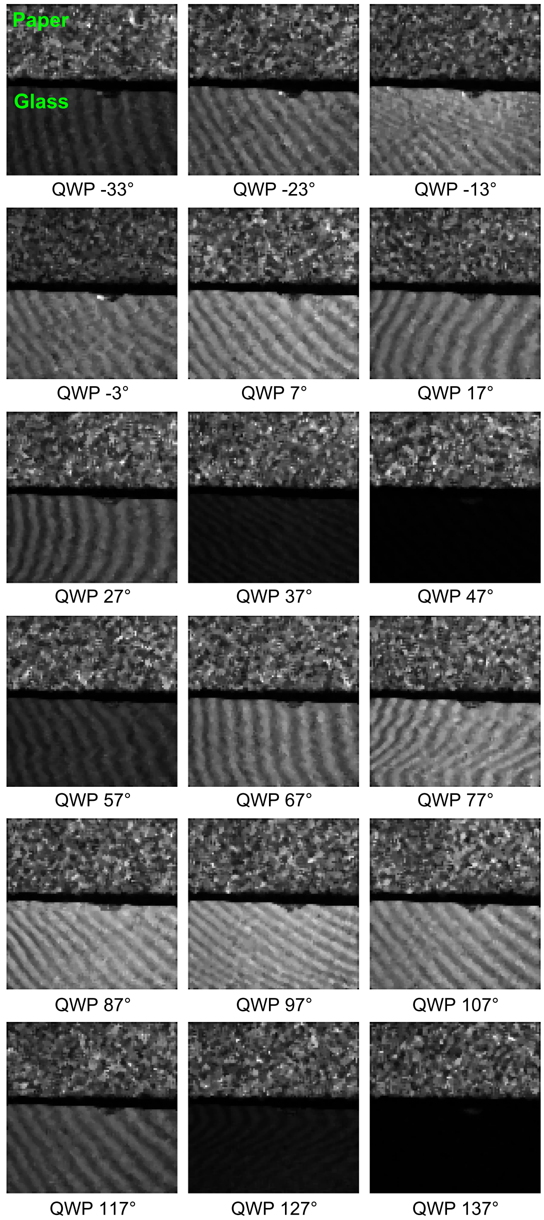

Fig. 4. Images of white paper (top) next to glass cover slip (bottom). QWP angle is swept from −33° to 137° (10° increments). Paper image does not depend on the QWP angle, whereas the glass image is strongly faded out at angles close to 45° and 135° where the specular reflection is filtered out.

This process generates rough areas with a roughness in the order of the wavelength of the VCSEL (850 nm) used in the polarization-sensitive LFI setup. Therefore, we expected the reflected beam from the acrylic and letter surface to be specular and diffuse (or at least partially diffuse), respectively. An area of 4.2 mm × 4.2 mm on the surface was scanned at two QWP angles of 0° and 45° relative to the incident linearly polarized beam.

Figures 5(a)–5(c) show photography of the engraved letters on the acrylic surface, LFI images of the letters with QWP at 0° and 45°, respectively. The background signal from the top glossy surface is strongly suppressed when the QWP angle is at 45°, whereas the signal from the engraved letters stays approximately at the same level. Part (d) of this figure shows the averaged LFI signal along the lines crossing the center of the images in part (b) and (c) (included as insets of blue and red broken lines). The higher level in the blue (QWP 0°) and the lower level in the red (QWP 45°) broken lines are associated with the top glossy surface, while the lower level in the blue (QWP 0°) and the higher level in the red (QWP 45°) broken lines are associated with the bottom matte surface. Strong suppression of the specular signal (top surface) is observed (longer arrow in Fig. 5(d) shows the specular suppression), while the drop in the diffuse signal (bottom surface) is relatively mild (shorter arrow in Fig. 5(d) shows the diffuse suppression), when the QWP angle switches from 0° to 45°. Higher fluctuation in the signal from engraved areas is as a result of the surface roughness.

C. Flow Channel

Here we applied the polarization-sensitive LFI system to image a microfluidic channel embedded in a silicone based layer. A 120 μm diameter cylindrical channel was built at a depth of 50 μm from the surface of a 1 mm thick silicone layer composed of two-part polydimethylsiloxane (PDMS), a vulcanizing room temperature silicone. The channel contained Intralipid 20%, which is a turbid medium. Intralipid is an emulsion of nanosized fat particles with a high scattering property at 850 nm that acts as a diffuser of the beam close to the specular silicone surface. An area of 0.3 mm × 0.3 mm in 61 × 61 steps at 5 μm pitch on the sample was scanned with the QWP angle at 0° and 45° to the incident linearly polarized beam. Figures 6(a) and 6(b) show the images of the microfluidic channel at depths 0 μm (on the surface) and 100 μm from the surface of the sample, respectively. Columns (i) and (ii) in Fig. 6 correspond to the QWP angle placed at 0° and 45° with respect to the incident linearly polarized beam, respectively. In Fig. 6, a two-dimensional median filter is applied to the images and each normalized to the maximum level of signal over the scan area. The reflected beam from the microfluidic sample had two components: a specular component, which originated from the surface due to the refractive index mismatch, and a diffuse component as a result of beam interaction with the Intralipid. Reflection of the photons on the surface is a single scattering event, but their interaction with Intralipid is a multiple scattering event. Normally, a photon has to undergo a large number of scatterings to finally propagate in a reverse direction, and its polarization is scrambled during this multiple scattering process. As mentioned before, only the depolarized beam contributes to the signal when the QWP is at 45°. This is the case in Fig. 6(ii), whereas in Fig. 6(i) front surface reflection has covered the signal from within the channel; this is the part that contains the main information (about the contents of the channel). The sample in this experiment can be considered as a model for skin tissue with subsurface microvasculature, and the results demonstrated the effectiveness of the technique for such an application.

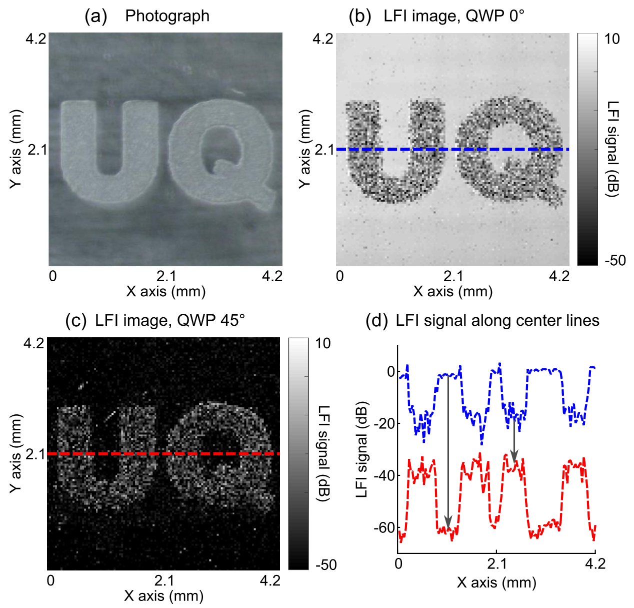

Fig. 5. Polarization-sensitive LFI images of laser engraved letters on acrylic surface, (a) a photography of the sample, (b) image with QWP at 0°, (c) image with QWP at 45° (with respect to the incident linearly polarized beam), and (d) LFI signal along the lines crossing the center of the images in part (b) and (c), shown as broken lines in blue (QWP at 0°) and red (QWP at 45°). The long and short arrows in part (d) indicate the drops in the specular and diffuse LFI signals when changing the QWP angle from 0° to 45°, respectively.

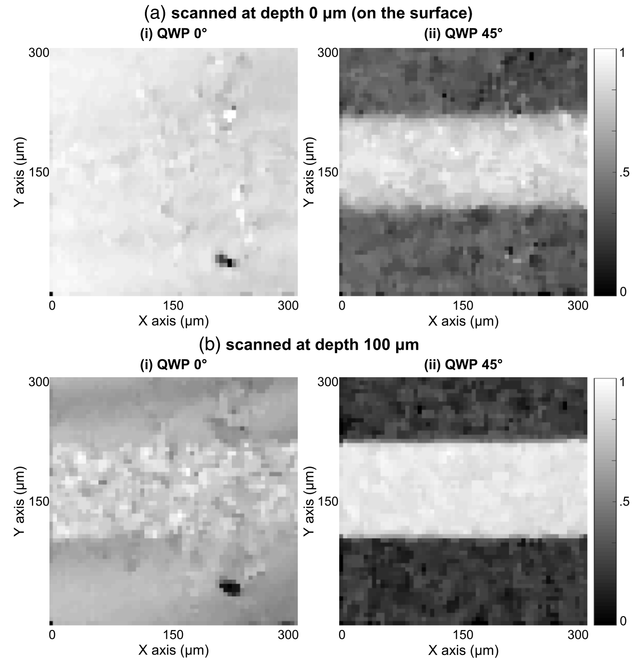

Fig. 6. 120 μm diameter microfluidic channel embedded in a silicone layer containing Intralipid 20% imaged at depths of (a) 0 μm (on the surface) and (b) 100 μm from the surface of the sample. The QWP angle is placed at 0°, column (i), and at 45°, column (ii), to the incident linearly polarized beam. Images have been normalized to their maximum values.

4. DISCUSSION

A laser is a coherent source of light, and in most cases it emits a polarized beam. The polarization state of a laser beam may experience changes in a sensing or imaging system as it passes though components or reflects off surfaces. Monitoring, manipulating, and selectively filtering out polarization states in so-called polarization-sensitive systems can dramatically improve their versatility. Introducing sensitivity to polarization in LFI systems with well-established biomedical applications [5,18] can have a great impact on their applicability in this field. For instance, when light is incident upon the surface of biological tissue, a portion of the beam will be reflected off the surface due to refractive index mismatch. This part mostly retains its polarization state. However a fraction of the beam power enters the tissue, where it will be scattered or absorbed by the structures within the tissue. Ultimately, a minute portion of the light power finds its way back to the surface due to multiple scattering and change of direction. This part is depolarized but contains information about the biomarkers within the tissue. Although more important, this part is extremely small compared to the specular part and will be completely covered by it. Here, a polarization-sensitive LFI system can exploit the differences in the polarization state of the two components to separate them and remove the specular part.

Light from noninvasive optical systems usually enters the tissue from outermost layer of the skin (epidermis) or a mucous membrane in the case of body internal cavities in an endoscopic system, both consisting of epithelial cell layers. Usually, mucous membrane is masked with mucus making the surface more specular, while the dermis surface has a more diffuse behavior. Owing to the refractive index mismatch between air and such surfaces, light experiences a certain level of reflectance, in the order of a few percent of the incident light intensity, while passing through. Within the tissue, the dominant interaction is the scattering events as the refractive index mismatch is much lower (about one order of magnitude lower) at the internal interfaces such as blood-tissue boundaries. In the experiments conducted in this work, light encountered a low level of reflectance at the PDMS-Intralipid interface (due to the low level of refractive index mismatch), while the reflectance was higher at air-PDMS and air-acrylic interfaces. The refractive indices of PDMS, Intralipid, and acrylic are about 1.4 while the value for air is one.

As the photons propagate through a biological tissue, the number of photons that remain unscattered decreases exponentially [24]. A typical value for photon mean free path (the average distance between two consecutive scattering events) in a biological tissue is about 100 μm [25]. Complete depolarization occurs when photons are scattered at least 10 times [10] in most cases. Therefore, depth of travel of a polarized photon in a biological tissue can also be associated with the degree of depolarization. Based on this fact, depth resolved biomedical imaging can also be investigated using a polarization-sensitive system such as our proposed technique.

An ideal diffuse surface has a Lambertian profile of reflectance and completely depolarizes an incident polarized beam. This type of ideal surface is rare in practical situations as photons should be scattered uniformly numerous times upon reflection. The LFI signal from such a surface is expected to have a flatter profile against a rotating QWP angle (Fig. 3). In addition to Spectralon, which is a commercially available diffuse standard, we tried other diffuse surfaces such as white paper, tape with titanium dioxide powder on it, and unfinished sandpaper-polished wood. We also found wood as a relatively good diffuser of the beam. Wood has a porous structure consisting mostly of cellulose fibers. Light penetrates the wood slightly and reflects back diffusely after multiple scattering from the internal structure. In the case of reflection from rough surfaces, scattering happens on the surface with no beam penetration. In most cases the beam is only partially diffused with enhanced backscattering of the beam [26]. Moreover, surface roughness is a concept with regard to the beam's wavelength, as the size of the random structure on the surface should be of the same order of magnitude with the beam's wavelength. To create a rough surface in Section 3.B, a CO_2 laser was used to engrave letters on an acrylic surface. Laser power evaporated the plastic and generated random structures on the surface. We examined the letter surfaces under a microscope and confirm that the roughness was of the order of a few micrometers, rough enough to partially diffuse a VCSEL's wavelength of 850 nm.

Considering the collimating and focusing lens arrangement in this setup, we estimate the $1/e^2$ focal beam spot diameter to be about 10 mm. This beam spot size is small enough for the imaging applications in this work, and at the same time it allows us to experience a probing beam with a relatively long depth of field and still be able to effectively suppress the front surface reflection and acquire the signal from a shallow depth beneath the surface (to verify a relaxed depth of field constraint in this technique). If a higher resolution is desired, for instance in a microscopy application, we can replace the focusing lens with a higher NA lens [27]. Furthermore, in the course of the experiments, different types of targets were used that incorporated different levels of feedback. For instance, the reflection from specular surfaces were much stronger than the diffuse reflection from the embedded Intralipid channel. Therefore, different regimes of feedback were expected (strong feedback from the specular surfaces and moderate to weak feedback from the diffuse targets). To extract the signal, a laser beam was modulated using an optical chopper, and the shape of the interferometric fringes was irrelevant. As a result, two levels of signal were obtained that were directly proportional to the level of the feedback from the targets.

5. CONCLUSION

We proposed a polarization-sensitive laser feedback interferometry system for sensing and imaging applications. This system is capable of controlling the polarization state of the incident beam and distinguishing between the polarized and depolarized parts of the reflected beam. This can be of great importance in applications such as biomedical imaging, to filter out an unwanted part of the signal that has a different polarization due to a different mechanism of reflection and let the weaker part that contains information through. The idea has been studied theoretically and examined experimentally, and applications such as front surface specularity removal have been investigated. We believe this simple configuration can be added to the previously existing laser feedback interferometry systems to improve their versatility, in applications such as high-contract and depth-resolved biomedical imaging and compound target characterization.

APPENDIX A: STOKES PARAMETERS AND MUELLER MATRIX FORMULATION

In general, the polarization state of a wave and its conversion upon reflection from different types of surfaces and passage through transmissive optical components such as polarizers and wave plates can be described by means of Stokes matrix formalism, using Stokes vector as the beam polarization state and Mueller matrix as a linear transformation [28]. Stokes parameters are a set of values that relate directly to the measured intensities of different polarizations and can be combined to make a Stokes vector. The 4 × 4 Mueller matrix is an operator or a transformation matrix, associated with an optical device, relating the input and output Stokes parameters.

In this appendix we present an analytical model for the proposed system based on Stokes vectors and Mueller matrix formulation. This model explains the theory of polarization conversion throughout the system by means of Mueller calculus, which is a matrix method. In this model we consider Mueller matrices for all the components that affect the beam polarization in the setup. By multiplying all Mueller matrices together, an overall Mueller matrix for the whole system is calculated. Ultimately, the output Stokes vector, which is the polarization state of the beam returning into the VCSEL's cavity, is found by multiplying the input Stokes vector by the overall Mueller matrix. Some of the common Stokes vectors used in this calculation are [28]

$$S_{L_V} = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \quad S_{L_H} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad S_{\mathrm{RCP}} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$
$$S_{\mathrm{LCP}} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \quad S_{\mathrm{dep}} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{A1}$$

where S_{L_V}, S_{L_H}, S_{RCP}, S_{LCP}, and S_{dep} are the Stokes vectors for vertical linearly polarized, horizontal linearly polarized, RCP, LCP, and depolarized beams, respectively. Also, the required Mueller matrices for this calculation are [28]

$$M_{P_V} = \frac{1}{2}\begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad M_{\mathrm{QWP}_{+45^\circ}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix},$$
$$M_{\mathrm{QWP}_{-45^\circ}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad M_{\mathrm{spc}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix},$$
$$M_{\mathrm{diff}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \tag{A2}$$

where M_{P_V}, $M_{\mathrm{QWP}_{+45^\circ}}$, $M_{\mathrm{QWP}_{-45^\circ}}$, M_{spc}, and M_{diff} are the Mueller matrices for vertical polarizer, QWP at +45°, QWP at −45°, specular, and diffuse surfaces, respectively.

Ignoring the components that do not change the polarization state of the beam (the optical chopper and lenses), the interaction of the VCSEL beam with the system is as follows: the beam passes through the polarizer and QWP, reflects off the target, and passes through the QWP and polarizer again on its way back (see Fig. 2). The vertical polarizer has the same Mueller matrix for both the forward and backward propagating beams, whereas the QWP at +45° with respect to the forward beam is at −45° with respect to the backward beam resulting in two different Mueller matrices as described in Section 2. Considering all Mueller matrices, cascading of polarizing elements transform the input Stokes vectors into the output Stokes vectors (for the specular and diffuse surfaces),

$$S_{\mathrm{out_{spc}}} = M_{P_V} M_{\mathrm{QWP}_{-45^\circ}} M_{\mathrm{spc}} M_{\mathrm{QWP}_{+45^\circ}} M_{P_V} S_{L_V} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{A3}$$

and

$$S_{\mathrm{out_{diff}}} = M_{P_V} M_{\mathrm{QWP}_{-45^\circ}} M_{\mathrm{diff}} M_{\mathrm{QWP}_{+45^\circ}} M_{P_V} S_{L_V} = \frac{1}{2}\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \tag{A4}$$

where $S_{\mathrm{out_{spc}}}$ and $S_{\mathrm{out_{diff}}}$ are the output Stokes vectors for the specular and diffuse surfaces, respectively. The resulting values of $S_{\mathrm{out_{spc}}}$ and $S_{\mathrm{out_{diff}}}$ equate to no beam and a half-sized vertical linearly polarized beam. It can be concluded from the Mueller calculation that specular reflection does not find its way back into the laser cavity and does not contribute to the signal when QWP is at 45°. However, diffuse reflection is able to pass through the system and contribute to the signal (although half of the signal is blocked). Attenuation upon reflection was not accounted for in this model, although it may affect the level of signal especially in the case of diffuse reflection. However, it does not interfere with the polarization states of the beam.

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REFERENCES

1. G. Giuliani, M. Norgia, S. Donati, and T. Bosch, "Laser diode self-mixing technique for sensing applications," J. Opt. A **4**, S283–S294 (2002).
2. S. Donati, "Developing self-mixing interferometry for instrumentation and measurements," Laser Photon. Rev. **6**, 393–417 (2012).
3. T. Taimre, M. Nikolić, K. Bertling, Y. L. Lim, T. Bosch, and A. D. Rakić, "Laser feedback interferometry: a tutorial on the self-mixing effect for coherent sensing," Adv. Opt. Photon. **7**, 570–631 (2015).
4. Y. L. Lim, T. Taimre, K. Bertling, P. Dean, D. Indjin, A. Valavanis, S. P. Khanna, M. Lachab, H. Schaider, T. W. Prow, H. P. Soyer, S. J. Wilson, E. H. Linfield, A. G. Davies, and A. D. Rakić, "High-contrast coherent terahertz imaging of porcine tissue via swept-frequency feedback interferometry," Biomed. Opt. Express **5**, 3981–3989 (2014).

5. S. Donati and M. Norgia, "Self-mixing interferometry for biomedical signals sensing," IEEE J. Sel. Top. Quantum Electron. **20**, 104–111 (2014).
6. J. Perchoux, A. Quotb, R. Atashkhooei, F. J. Azcona, E. E. Ramírez-Miquet, O. Bernal, A. Jha, A. Luna-Arriaga, C. Yanez, J. Caum, T. Bosch, and S. Royo, "Current developments on optical feedback interferometry as an all-optical sensor for biomedical applications," Sensors **16**, 694 (2016).
7. A. Mowla, B. W. Du, T. Taimre, K. Bertling, S. Wilson, H. P. Soyer, and A. D. Rakić, "Confocal laser feedback tomography for skin cancer detection," Biomed. Opt. Express **8**, 4037–4048 (2017).
8. R. R. Anderson, "Polarized light examination and photography of the skin," Arch. Dermatol. **127**, 1000–1005 (1991).
9. S. L. Jacques, J. C. Ramella-Roman, and K. Lee, "Imaging skin pathology with polarized light," J. Biomed. Opt. **7**, 329–340 (2002).
10. W. Groner, J. W. Winkelman, A. G. Harris, C. Ince, G. J. Bouma, K. Messmer, and R. G. Nadeau, "Orthogonal polarization spectral imaging: a new method for study of the microcirculation," Nat. Med. **5**, 1209–1212 (1999).
11. J. F. De Boer, S. M. Srinivas, A. Malekafzali, Z. Chen, and J. S. Nelson, "Imaging thermally damaged tissue by polarization sensitive optical coherence tomography," Opt. Express **3**, 212–218 (1998).
12. S. Zhang and W. Holzapfel, *Orthogonal Polarization in Lasers: Physical Phenomena and Engineering Applications* (Wiley, 2013).
13. W. Mao, S. Zhang, L. Cui, and Y. Tan, "Self-mixing interference effects with a folding feedback cavity in Zeeman-birefringence dual frequency laser," Opt. Express **14**, 182–189 (2006).
14. P. Porta, D. Curtin, and J. McInerney, "Laser Doppler velocimetry by optical self-mixing in vertical-cavity surface-emitting lasers," IEEE Photon. Technol. Lett. **14**, 1719–1721 (2002).
15. O. Jacquin, S. Heidmann, E. Lacot, and O. Hugon, "Self-aligned setup for laser optical feedback imaging insensitive to parasitic optical feedback," Appl. Opt. **48**, 64–68 (2009).
16. L. Campagnolo, M. Nikolić, J. Perchoux, Y. L. Lim, K. Bertling, K. Loubiere, L. Prat, A. D. Rakić, and T. Bosch, "Flow profile measurement in microchannel using the optical feedback interferometry sensing technique," Microfluid. Nanofluid. **14**, 113–119 (2013).
17. F. Mezzapesa, L. Columbo, M. Dabbicco, M. Brambilla, and G. Scamarcio, "QCL-based nonlinear sensing of independent targets dynamics," Opt. Express **22**, 5867–5874 (2014).
18. A. Mowla, T. Taimre, Y. L. Lim, K. Bertling, S. J. Wilson, T. W. Prow, H. P. Soyer, and A. D. Rakić, "Concurrent reflectance confocal microscopy and laser Doppler flowmetry to improve skin cancer imaging: a Monte Carlo model and experimental validation," Sensors **16**, 1411 (2016).
19. J. Keeley, J. Freeman, K. Bertling, Y. L. Lim, R. A. Mohandas, T. Taimre, L. H. Li, D. Indjin, A. D. Rakić, E. H. Linfield, A. G. Davies, and P. Dean, "Measurement of the emission spectrum of a semiconductor laser using laser-feedback interferometry," Sci. Rep. **7**, 7236 (2017).
20. K. Bertling, J. Perchoux, T. Taimre, R. Malkin, D. Robert, A. D. Rakić, and T. Bosch, "Imaging of acoustic fields using optical feedback interferometry," Opt. Express **22**, 30346–30356 (2014).
21. A. Mowla, T. Taimre, Y. L. Lim, K. Bertling, S. J. Wilson, T. W. Prow, and A. D. Rakić, "A compact laser imaging system for concurrent reflectance confocal microscopy and laser Doppler flowmetry," IEEE Photon. J. **8**, 1–9 (2016).
22. R. S. Matharu, J. Perchoux, R. Kliese, Y. L. Lim, and A. D. Rakić, "Maintaining maximum signal-to-noise ratio in uncooled vertical-cavity surface-emitting laser-based self-mixing sensors," Opt. Lett. **36**, 3690–3692 (2011).
23. K. Bertling, T. Taimre, G. Agnew, Y. L. Lim, P. Dean, D. Indjin, S. Höfling, R. Weih, M. Kamp, M. von Edlinger, J. Koeth, and A. D. Rakić, "Simple electrical modulation scheme for laser feedback imaging," IEEE Sens. J. **16**, 1937–1942 (2016).
24. S. G. Demos, H. B. Radousky, and R. R. Alfano, "Deep subsurface imaging in tissues using spectral and polarization filtering," Opt. Express **7**, 23–28 (2000).
25. L. Wang, S. L. Jacques, and L. Zheng, "MCML-Monte Carlo modeling of light transport in multi-layered tissues," Comput. Methods Programs Biomed. **47**, 131–146 (1995).
26. A. Mowla, M. Nikolić, T. Taimre, J. R. Tucker, Y. L. Lim, K. Bertling, and A. D. Rakić, "Effect of the optical system on the Doppler spectrum in laser-feedback interferometry," Appl. Opt. **54**, 18–26 (2015).
27. A. Mowla, T. Taimre, K. Bertling, S. J. Wilson, and A. D. Rakić, "Confocal laser feedback microscopy for in-depth imaging applications," Electron. Lett. **54**, 196–198 (2018).
28. H. Tompkins and E. A. Irene, *Handbook of Ellipsometry* (William Andrew, 2005).